

GAINESVILLE, TX, USA

WMO: 722552

Lat: 33.651N

Lon: 97.197W

Elev: 256

StdP: 98.29

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 93929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-4.6	-15.4	1.0	-1.4	-12.6	1.3	2.5	12.1	8.5	11.1	10.4	3.2	340	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	37.8	22.4	37.1	22.7	35.1	23.3	25.8	32.7	25.3	32.5	24.9	31.9	4.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.4	29.5	23.0	18.4	28.0	22.8	18.1	27.8	81.3	32.6	78.9	32.8	76.8	31.6	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.3	8.4	-10.8	39.5	2.2	2.5	-12.4	41.3	-13.7	42.8	-14.9	44.1	-16.5	45.9		
			-11.8	27.2	1.9	0.9	-13.2	27.8	-14.2	28.4	-15.3	28.9	-16.6	29.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	6.4	8.3	13.5	17.1	21.7	26.5	28.5	28.7	24.7	18.4	12.6	7.1
	DBStd	9.07	5.49	6.04	5.42	4.32	3.91	2.58	2.38	2.71	3.23	4.81	5.09	5.15
	HDD10.0	421	139	95	29	5	0	0	0	0	0	3	31	119
	HDD18.3	1494	372	282	164	72	15	0	0	0	1	59	180	349
	CDD10.0	3281	26	48	137	217	362	494	572	581	441	263	110	29
	CDD18.3	1312	1	3	15	33	118	244	314	322	192	61	9	1
	CDH23.3	14430	7	25	105	246	1005	2608	3887	4016	1911	541	72	7
	CDH26.7	7132	1	3	16	43	345	1264	2157	2254	872	164	13	0
Wind	WSAvg	3.6	3.7	4.1	4.4	4.5	3.9	3.5	3.0	2.8	2.8	3.3	3.7	3.6
Precipitation	PrecAvg	1010	51	66	93	105	112	109	65	65	95	102	69	74
	PrecMax	1360	116	202	204	306	236	405	209	203	277	288	186	216
	PrecMin	574	0	2	7	37	12	17	0	2	20	5	9	7
	PrecStd	221	27	49	51	68	55	84	55	45	61	75	57	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	26.2	28.1	31.1	34.1	37.3	40.9	42.1	37.2	32.4	28.1	24.0
		MCWB	12.8	16.1	18.3	19.0	22.8	23.1	22.4	22.1	22.1	21.3	18.2	15.1
	2%	DB	21.3	23.0	26.2	27.4	32.3	35.2	37.7	38.0	34.8	31.0	25.8	21.0
		MCWB	13.6	15.3	17.9	19.2	22.7	23.4	22.2	22.5	22.2	21.4	17.1	14.0
	5%	DB	18.5	20.9	23.9	26.1	30.9	34.0	37.0	37.3	32.9	28.2	22.9	17.9
		MCWB	11.6	14.8	16.7	18.6	22.9	23.1	22.7	22.8	22.2	19.9	16.4	12.6
	10%	DB	15.9	18.0	22.5	24.2	28.3	32.8	35.0	35.3	31.7	26.4	21.1	15.8
		MCWB	10.2	13.3	16.3	17.7	21.5	23.8	23.5	23.6	22.0	18.7	15.7	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	19.2	21.6	22.9	25.7	26.8	26.6	26.4	25.6	24.0	20.9	18.6
		MCDB	20.1	23.1	24.9	26.9	32.1	33.2	33.4	32.8	32.3	29.5	25.2	22.0
	2%	WB	15.3	17.8	20.0	21.8	24.5	25.7	25.7	25.6	24.6	22.9	19.5	16.3
		MCDB	18.1	21.1	23.3	25.3	30.1	32.4	32.8	32.9	31.0	28.0	23.1	19.0
	5%	WB	13.0	16.1	19.0	20.6	23.4	25.1	25.2	25.1	23.9	22.0	18.2	14.2
		MCDB	17.1	19.0	22.8	24.0	28.1	31.7	32.3	32.7	29.9	26.0	21.6	16.4
	10%	WB	10.5	14.0	17.7	19.5	22.5	24.4	24.6	24.6	23.3	21.0	16.7	11.8
		MCDB	14.9	17.9	21.4	23.3	26.8	30.7	31.7	31.9	28.8	24.8	20.4	14.6
Mean Daily Temperature Range	5% DB	MDBR	12.3	11.7	12.0	11.7	10.7	10.7	11.6	11.7	11.6	12.2	12.2	11.4
		MCDBR	17.0	15.8	13.9	12.9	12.7	12.1	13.8	13.8	13.3	13.6	14.4	15.4
		MCWBR	10.3	9.6	7.8	7.3	5.8	4.4	3.8	3.7	4.7	6.4	8.4	9.6
	5% WB	MCDBR	14.2	13.3	10.8	10.3	10.5	10.8	11.3	11.7	10.7	10.6	11.8	11.9
		MCWBR	10.6	9.7	7.3	7.3	5.9	4.9	4.3	4.0	4.5	5.9	8.8	9.4
Clear-Sky Solar Irradiance	taub	0.307	0.317	0.351	0.381	0.419	0.432	0.434	0.433	0.392	0.352	0.326	0.311	
	taud	2.498	2.482	2.408	2.346	2.304	2.306	2.336	2.318	2.382	2.479	2.490	2.488	
	Ebn at Noon	910	937	925	906	871	855	851	847	871	885	878	883	
	Edh at Noon	85	96	112	125	131	131	126	127	114	95	85	81	
All-Sky Solar Radiation	RadAvg	2.85	3.39	4.46	5.43	5.86	6.60	6.71	6.16	5.13	3.99	3.03	2.48	
	RadStd	0.29	0.52	0.32	0.44	0.60	0.65	0.43	0.37	0.49	0.59	0.39	0.30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page